EC200U&EG91xU Series QuecOpen USBnet Call API Reference Manual

LTE Standard Module Series

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About the Document

Revision History

Version	Date	Author	Description
-	2021-12-09	Baron ZHENG	Creation of the document
1.0	2022-01-07	Baron ZHENG/ Braden HE	First official release
1.1	2023-08-17	Joe TU	Added the applicable modules: EG912U-GL and EG915U series.

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1 Introduction

Quectel EC200U and EG91xU family modules support QuecOpen® solution. QuecOpen® is an embedded development platform based on RTOS, which is intended to simplify the design and development of IoT applications. For more information on QuecOpen®, see ***document [1]***.

This document explains the USBnet call functionality, including API calling process, USBnet API and the example of Quectel EC200U and EG91xU family modules in QuecOpen® solution.

1.1. Applicable Modules

Table 1: Applicable Modules

Module Family	Module
-	EC200U Series
EG91xU	EG912U-GL
	EG915U Series

2 USBnet Calling Process

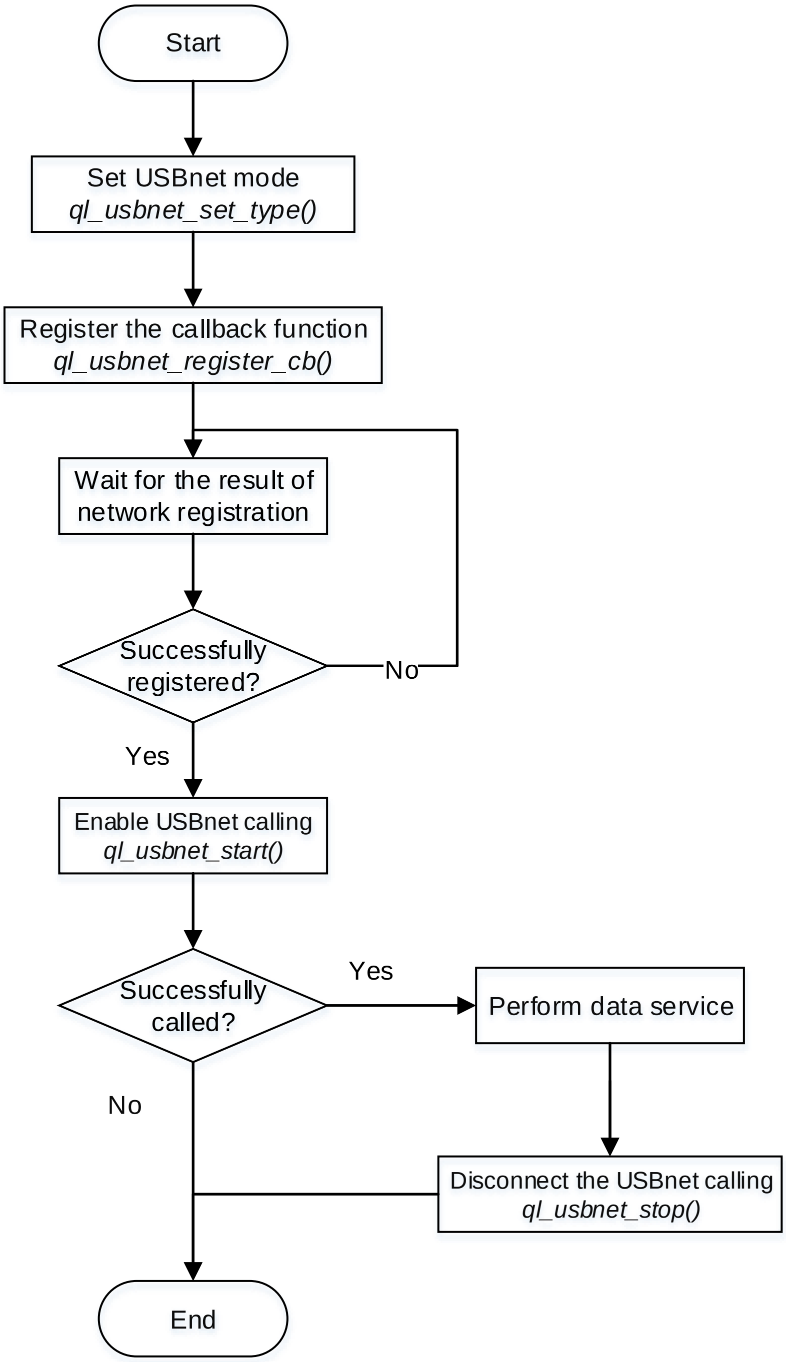

Figure 1: USBnet Calling Process

NOTE

Network registration is finished automatically without code intervention. You just need to call *ql_network_register_wait()* to wait for the completion of network registration. You can also check if the network registration is successful by calling *ql_nw_register_cb()* to register the network event callback function. For details about *ql_nw_register_cb() and ql_network_register_wait(),* see ***documents [2]*** and ***[3]*** respectively.

3 USBnet Call API

3.1. Header File

ql_api_usbnet.h, the header file of USBnet call API, is in the *components\ql-kernel\inc* directory. Unless otherwise specified, the header files mentioned in this document are all located in this directory.

3.2. API Overview

Table 2: API Overview

Function	Description
ql_usbnet_set_type()	Sets USBnet mode.
ql_usbnet_get_type()	Gets the current USBnet mode.
ql_usbnet_start()	Starts a USBnet call.
ql_usbnet_stop()	Stops a USBnet call.
ql_usbnet_get_status()	Gets the connection status of a USBnet call.
ql_usbnet_register_cb()	Registers the USBnet call callback function.

3.3. API Description

3.3.1. ql_usbnet_set_type

This function sets USBnet mode and, currently, only RNDIS and ECM are supported. Once the function is called, it takes effect after the module is rebooted.

- **Prototype**

```
ql_usbnet_errcode_e ql_usbnet_set_type(ql_usbnet_type_e usbnet_type)
```

- **Parameter**

usbnet_type:
[In] USBnet mode to be set. See ***Chapter 3.3.1.1*** for details.

- **Return Value**

See ***Chapter 3.3.1.2*** for details.

3.3.1.1. ql_usbnet_type_e

The enumeration of USBnet modes is defined as below:

```
typedef enum
{
    QL_USBNET_NONE = 0,
    QL_USBNET_ECM,
    QL_USBNET_MBIM,
    QL_USBNET_RNDIS,
    QL_USBNET_MAX
}ql_usbnet_type_e;
```

- **Member**

Member	Description
QL_USBNET_ECM	ECM mode.
QL_USBNET_MBIM	MBIM mode (not supported).
QL_USBNET_RNDIS	RNDIS mode.

3.3.1.2. ql_usbnet_errcode_e

The result codes of USBnet call indicate if the function is executed successfully. The enumeration of result codes is defined as below:

```
typedef enum
{
    QL_USBNET_SUCCESS                   = 0,
    QL_USBNET_EXECUTE_ERR               = 1 | QL_USBNET_ERRCODE_BASE,
    QL_USBNET_MEM_ADDR_NULL_ERR,
    QL_USBNET_INVALID_PARAM_ERR,
    QL_USBNET_USB_NOT_CONNECT_ERR,
    QL_USBNET_PDP_ACTIVE_ERR            = 5 | QL_USBNET_ERRCODE_BASE,
    QL_USBNET_REPEAT_CONNECT_ERR,
    QL_USBNET_REPEAT_DISCONNECT_ERR,
}ql_usbnet_errcode_e;
```

- **Member**

Member	Description
QL_USBNET_SUCCESS	Successful execution.
QL_USBNET_EXECUTE_ERR	Failed execution.
QL_USBNET_MEM_ADDR_NULL_ERR	Parameter address is NULL.
QL_USBNET_INVALID_PARAM_ERR	Invalid parameter.
QL_USBNET_USB_NOT_CONNECT_ERR	Unconnected USB.
QL_USBNET_PDP_ACTIVE_ERR	Failed to activate PDP.
QL_USBNET_REPEAT_CONNECT_ERR	Connect repeatedly.
QL_USBNET_REPEAT_DISCONNECT_ERR	Disconnect repeatedly.

3.3.2. ql_usbnet_get_type

This function gets the current USBnet mode.

- **Prototype**

```
ql_usbnet_errcode_e ql_usbnet_get_type(ql_usbnet_type_e *usbnet_type)
```

- **Parameter**

usbnet_type:
[Out] USBnet mode. See ***Chapter 3.3.1.1*** for details.

- **Return Value**

See ***Chapter 3.3.1.2*** for details.

3.3.3. ql_usbnet_start

This function starts a USBnet call. *QL_USBNET_SUCCESS* returned after function execution does not indicate a successful establishment of a call, but only a successful function execution. The registered callback function notifies the client Application if the call has been established successfully with *QUEC_USBNET_START_RSP_IND*. See ***Chapter 3.3.6.1*** for details.

- **Prototype**

```
ql_usbnet_errcode_e ql_usbnet_start(uint8_t nSim, int profile_idx, ql_data_call_conf_s *config)
```

- **Parameter**

nSim:
[In] (U)SIM card in use. If the module only supports one (U)SIM interface, the parameter can be set to 0.
  0 (U)SIM card 1
  1 (U)SIM card 2

profile_idx:
[In] PDP context ID. Range: 1–7.

config:
[In] Data call configuration. See ***Chapter 3.3.3.1*** for details. If the parameter is NULL, use default configuration.

- **Return Value**

See ***Chapter 3.3.1.2*** for details.

3.3.3.1. ql_data_call_conf_s

The structure of data call configuration is defined as below:

```
typedef struct
{
        int ip_version;
        char apn_name[APN_LEN_MAX];
        char username[USERNAME_LEN_MAX];
        char password[PASSWORD_LEN_MAX];
        int auth_type;
}ql_data_call_conf_s;
```

- **Parameter**

Type	Parameter	Description
int	*ip_version*	IP type. *<u>1</u>* IPv4 *2* IPv6 *3* IPv4v6
char	*apn_name*	Access Point Name. Maximum length: 64 bytes.
char	*username*	Username. Maximum length: 64 bytes.
char	*password*	Password. Maximum length: 64 bytes.
int	*auth_type*	Authentication type. *<u>0</u>* No authentication protocols *1* PAP authentication protocol *2* CHAP authentication protocol

3.3.4. ql_usbnet_stop

This function stops a USBnet call. After the call is stopped, the USBnet switches back to initial status.

- **Prototype**

```
ql_usbnet_errcode_e ql_usbnet_stop(void)
```

- **Parameter**

None

- **Return Value**

See ***Chapter 3.3.1.2*** for details.

3.3.5. ql_usbnet_get_status

This function gets the connection status of a USBnet call.

- **Prototype**

```
ql_usbnet_errcode_e ql_usbnet_get_status(ql_usbnet_state_e *status)
```

- **Parameter**

status:
[Out] Connection status of a USBnet call. See ***Chapter 3.3.5.1*** for details.

- **Return Value**

See ***Chapter 3.3.1.2*** for details.

3.3.5.1. ql_usbnet_state_e

The enumeration of connection status of a USBnet call is defined as below:

```
typedef enum
{
    QL_USBNET_STATE_NONE = 0,
    QL_USBNET_STATE_START,
    QL_USBNET_STATE_CONNECT,
    QL_USBNET_STATE_PORT_DISCONNECT,
    QL_USBNET_STATE_MAX,
}ql_usbnet_state_e;
```

- **Member**

Member	Description
QL_USBNET_STATE_NONE	Initial status.
QL_USBNET_STATE_START	The USBnet is connecting.

QL_USBNET_STATE_CONNECT	Successful connection.
QL_USBNET_STATE_PORT_DISCONNECT	The USB port is not connected.

3.3.6. ql_usbnet_register_cb

This function registers the USBnet call callback function. The USBnet notifies the client Application of the event with the registered callback function.

- **Prototype**

```
ql_usbnet_errcode_e ql_usbnet_register_cb(ql_usbnet_callback usbnet_cb, void *ctx)
```

- **Parameter**

usbnet_cb:
[In] Callback function of USBnet call event. See ***Chapter 3.3.6.1*** for function definition.

ctx:
[In] Argument pointer of callback function.

- **Return Value**

See ***Chapter 3.3.1.2*** for details.

3.3.6.1. ql_usbnet_callback

This function is the callback function of USBnet call event.

- **Prototype**

```
typedef void (*ql_usbnet_callback)(unsigned int ind_type, ql_usbnet_errcode_e errcode, void *ctx)
```

- **Parameter**

ind_type:
[In] USBnet call event type.

QUEC_USBNET_START_RSP_IND	USBnet call start response event.
QUEC_USBNET_DEACTIVE_IND	USBnet disconnection event including network signal loss or network disconnection.
QUEC_USBNET_PORT_CONNECT_IND	USB port connection event.
QUEC_USBNET_PORT_DISCONNECT_IND	USB port disconnection event.

errcode:
[In] Function execution result codes. See ***Chapter 3.3.1.2*** for details.

ctx:
[In] Argument pointer of callback function.

- **Return Value**

None

4 Example

4.1. USBnet Calling Demo

usbnet_demo.c, the example file of USBnet call provided in the SDK of Quectel EC200U and EG91xU family QuecOpen modules, is in the *components\ql-application\usbnet* directory.

4.2. Demo Auto-Start

Uncomment *ql_usbnet_app_init()* in *ql_init_demo_thread()* to start the Demo. The Demo can start automatically at bootup after the compiled firmware is downloaded to the module.

```
#ifdef QL_APP_FEATURE_USBNET
    //ql_usbnet_app_init();
#endif

#ifdef QL_APP_FEATURE_SFTP
    //ql_sftp_app_init();
#endif

    ql_rtos_task_sleep_ms(1000); /*Chaos change: set to 1000 for the camera power on*/
    ql_rtos_task_delete(NULL);
} « end ql_init_demo_thread »
```

Figure 2: Demo Auto-Start

5 Appendix References

Table 3: Related Documents

Document Name
[1] Quectel_EC200U&EG91xU_Series_QuecOpen_CSDK_Quick_Start_Guide
[2] Quectel_EC200U&EG91xU_Series_QuecOpen_Cellular _Network_Information_API_Reference_Manual
[3] Quectel_EC200U&EG91xU_Series_QuecOpen_Data_Call_API_Reference_Manual

Table 4: Terms and Abbreviations

Abbreviation	Description
API	Application Programming Interface
CHAP	Challenge Handshake Authentication Protocol
ECM	Ethernet Networking Control Model
IPv4	Internet Protocol Version 4
IPv6	Internet Protocol Version 6
MBIM	Mobile Broadband Interface Model
PAP	Password Authentication Protocol
PDP	Packet Data Protocol
RNDIS	Remote Network Driver Interface Specification
SDK	Software Development Kit
IoT	Internet of Things
USB	Universal Serial Bus

(U)SIM	(Universal) Subscriber Identity Module